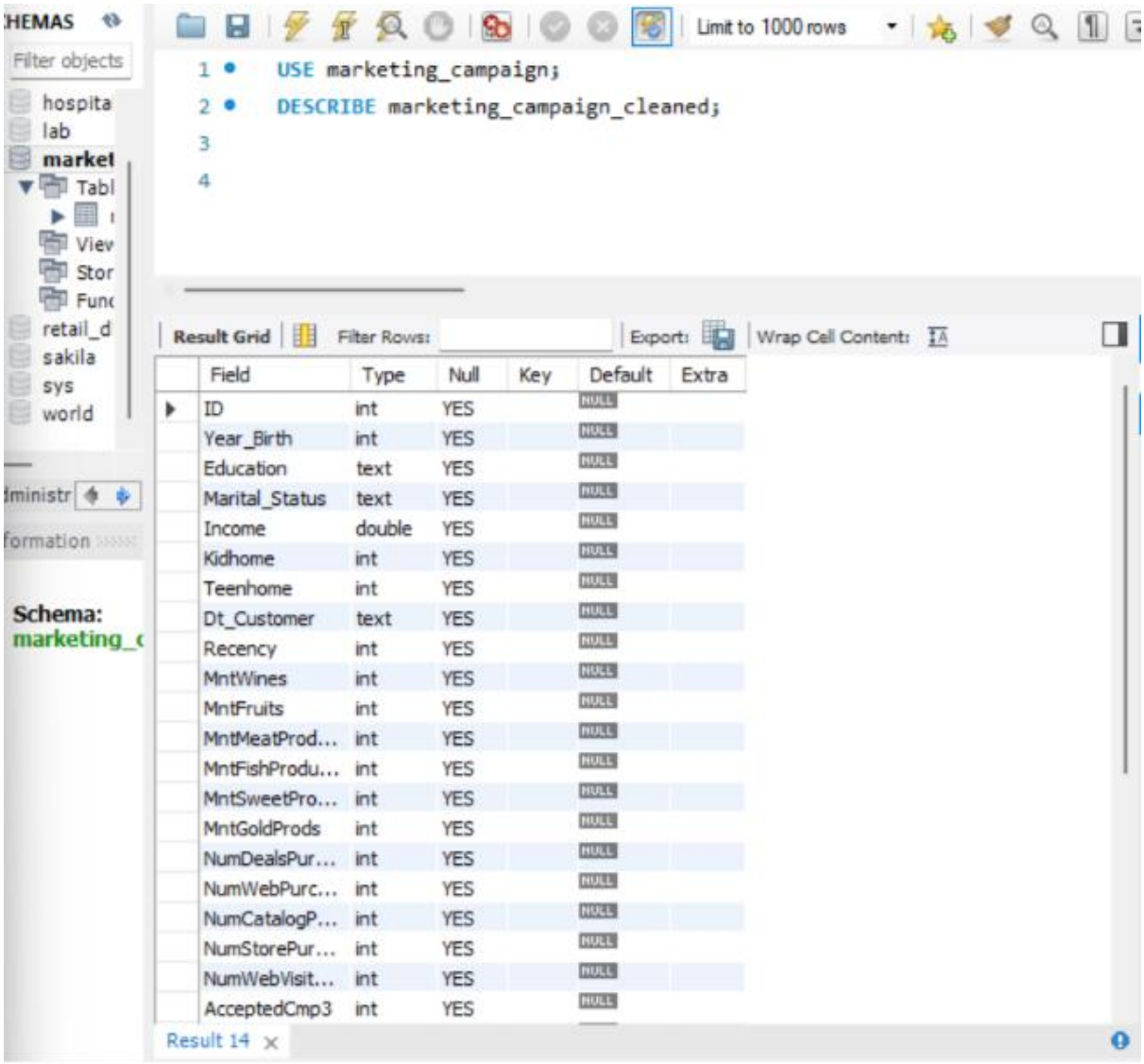
Database Schema

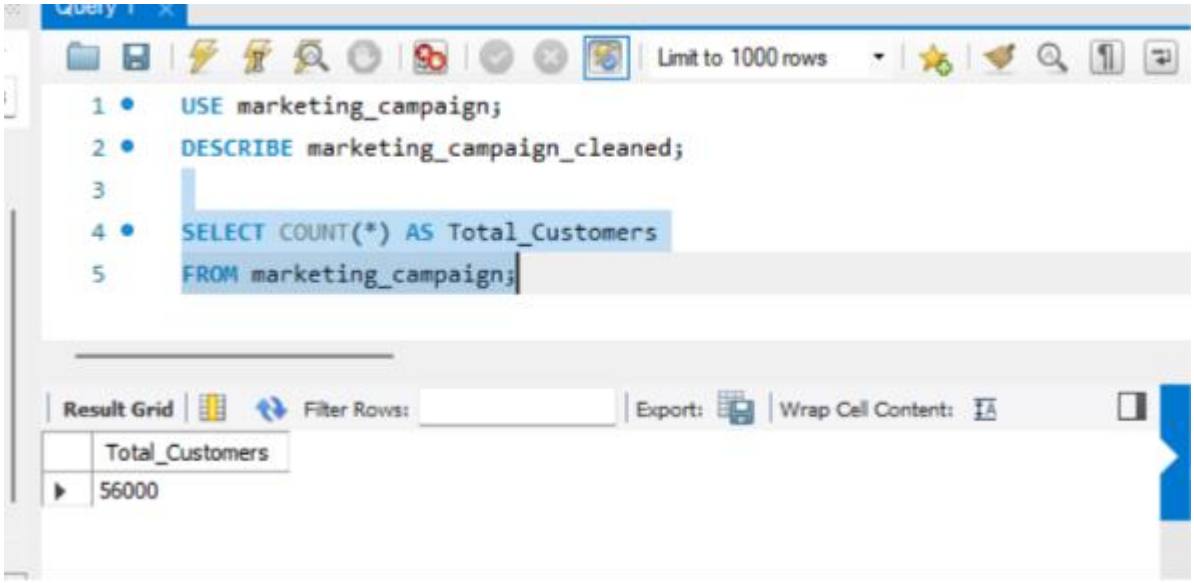


Report Explanation

The cleaned marketing campaign dataset was imported into MySQL. The table contains customer demographic information, purchase history, campaign response, and derived features such as Age, Customer_Tenure, Children, Total_Spend, and Total_Purchases.

KPI Analysis

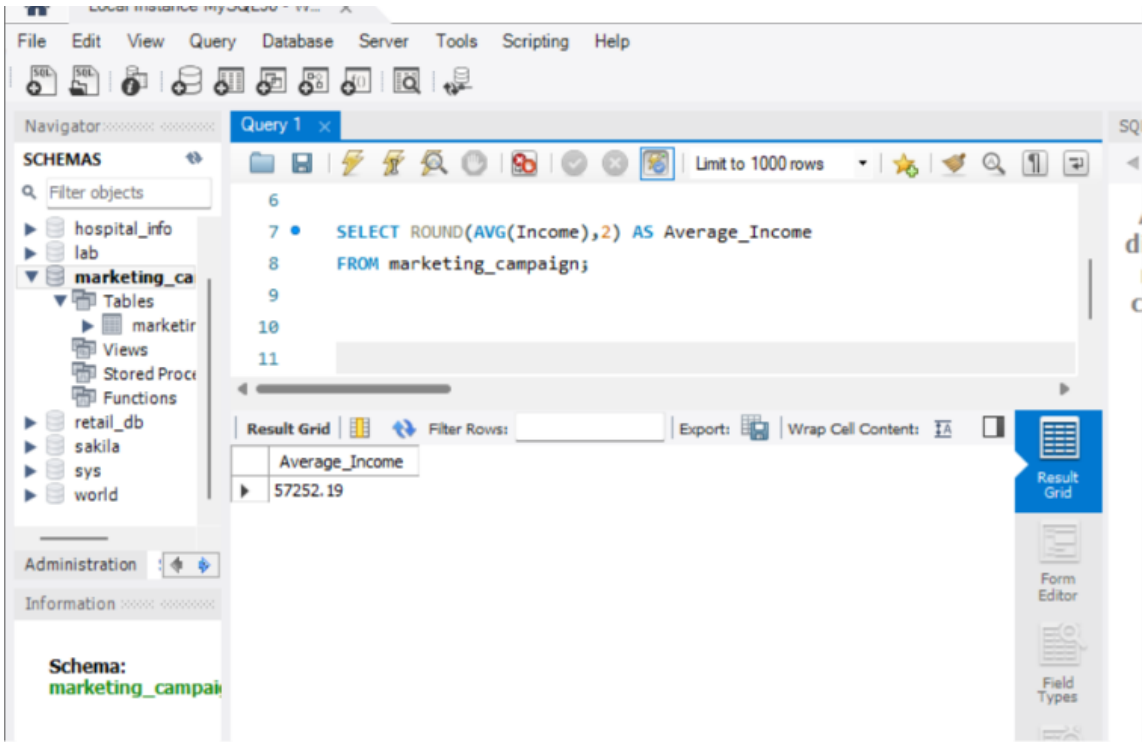
Query 1 –



Explanation:

This query calculates the total number of customer records available in the dataset.

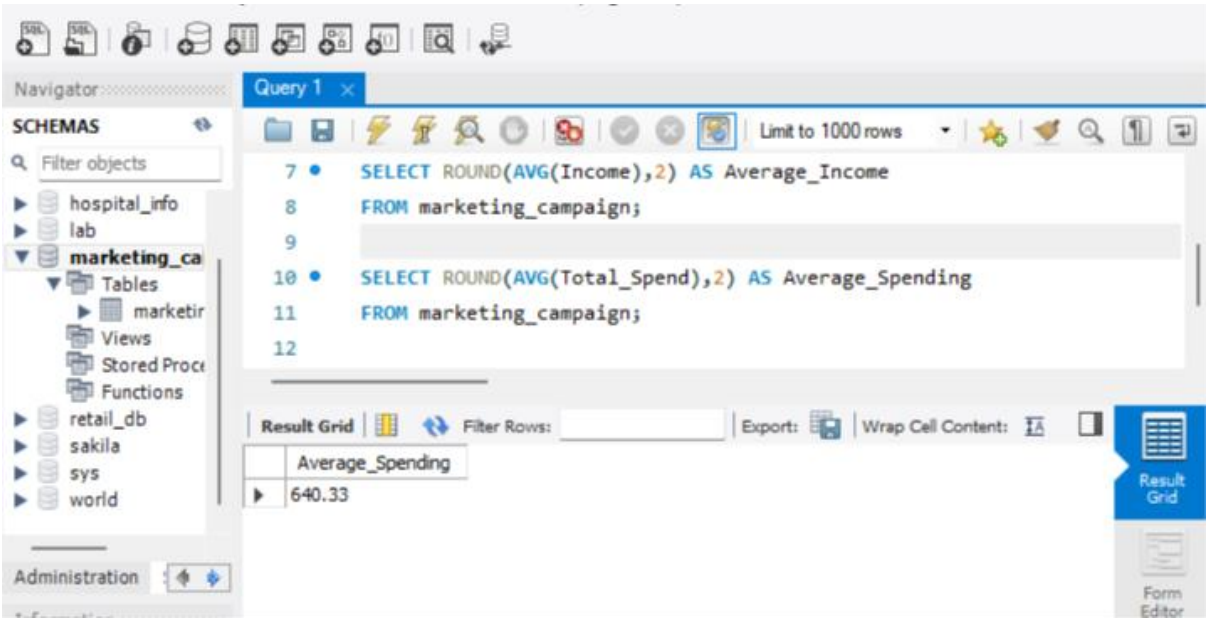
Query 2-



Explanation:

This query calculates the average annual income of customers.

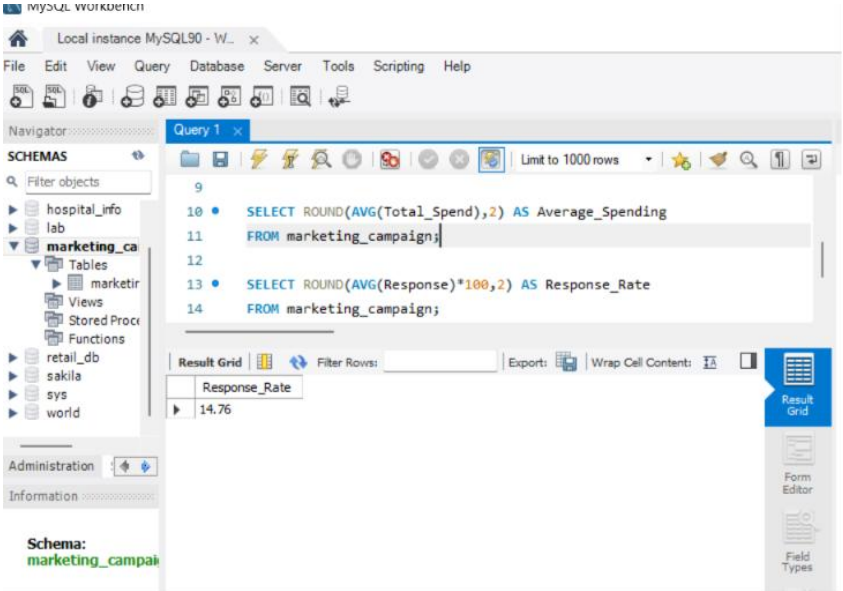
Query 3-



Explanation:

This query calculates the average spending of customers across all product categories.

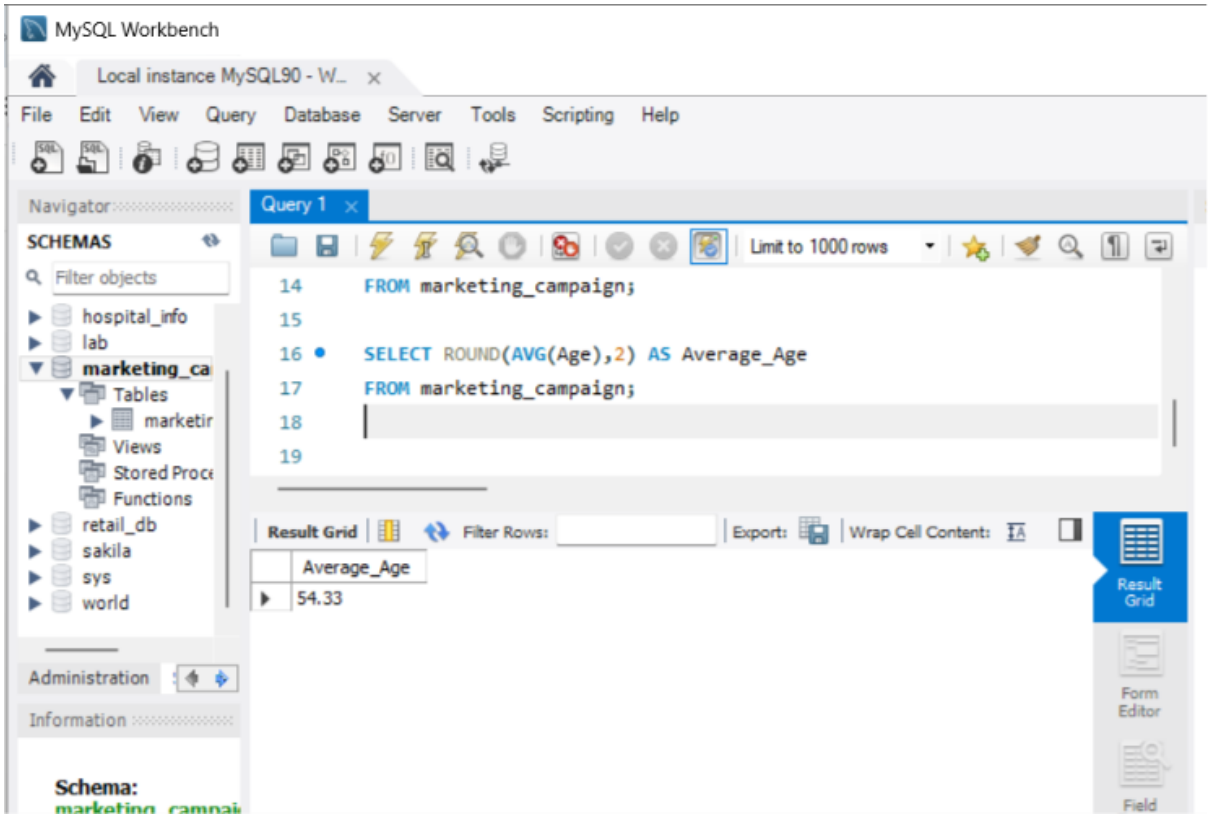
Query 4-



Explanation:

This query calculates the percentage of customers who responded positively to the marketing campaign.

Query 5 –

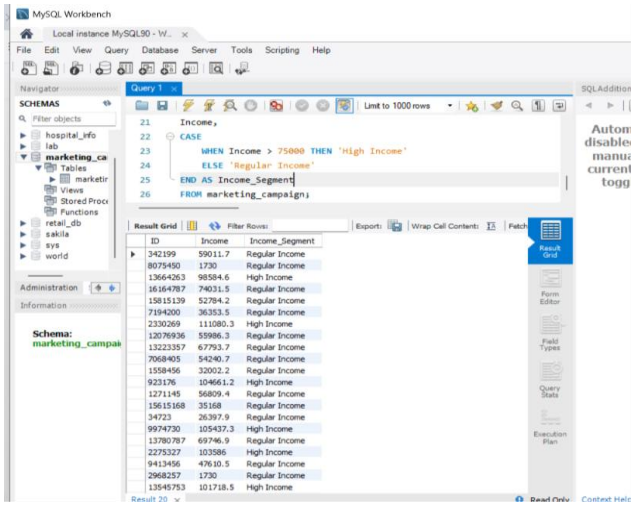


Explanation:

This query calculates the average age of customers.

Rule-Based Segmentation

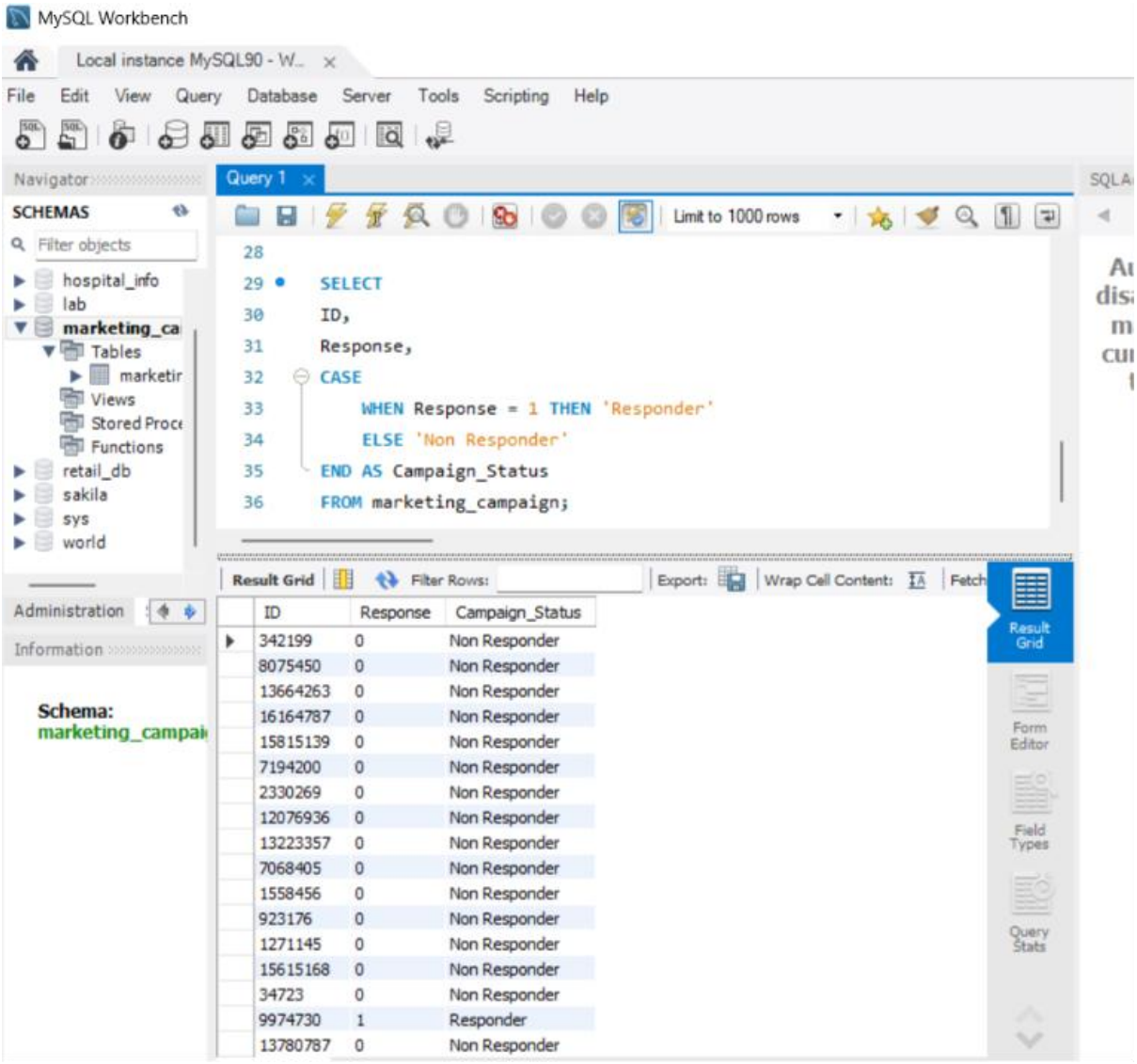
Query 1-High Income Customers



Explanation

This query classifies customers into **High Income** and **Regular Income** segments based on their annual income.

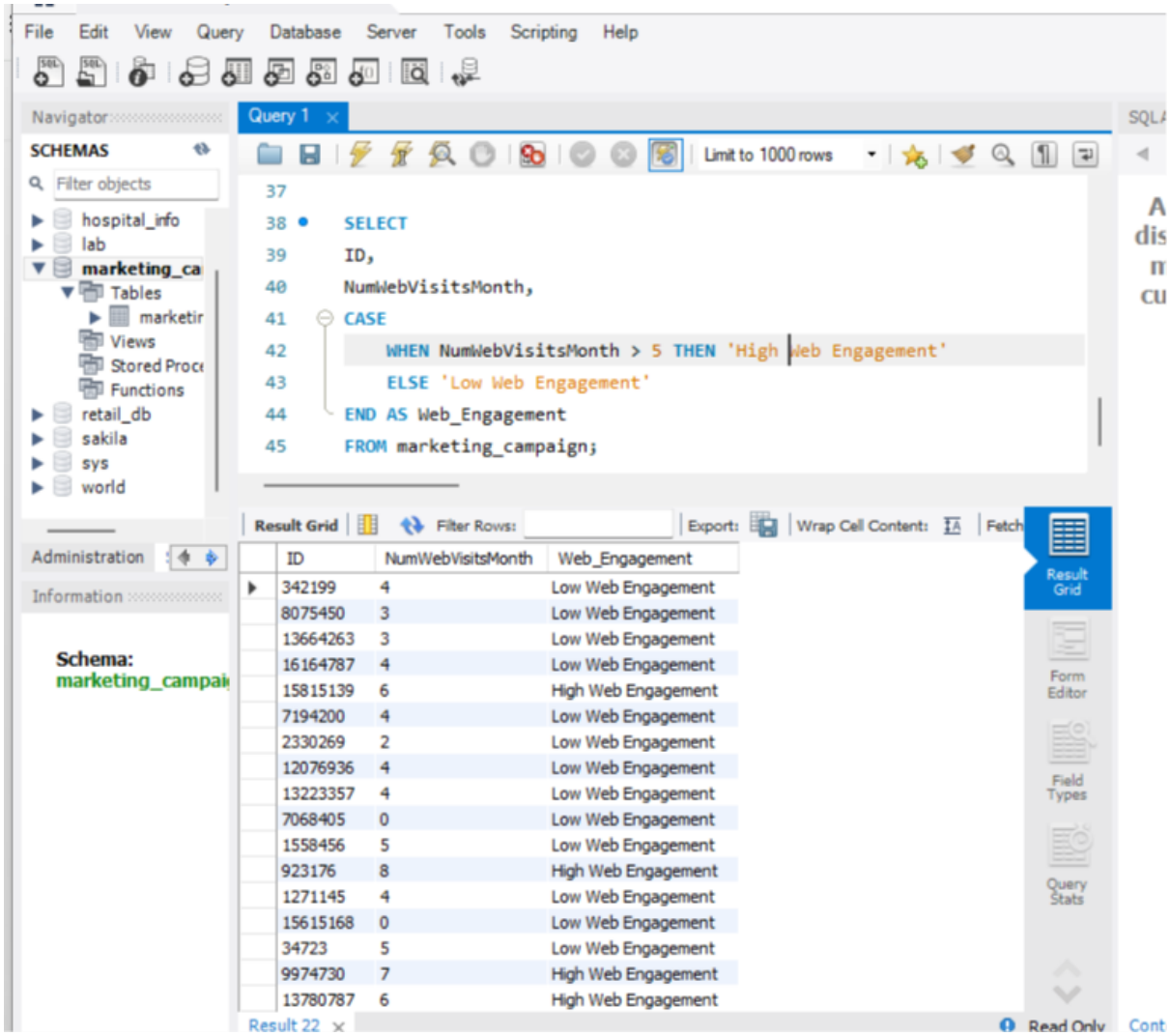
Query 3: Campaign Responders



Explanation

This query identifies customers who responded to the marketing campaign.

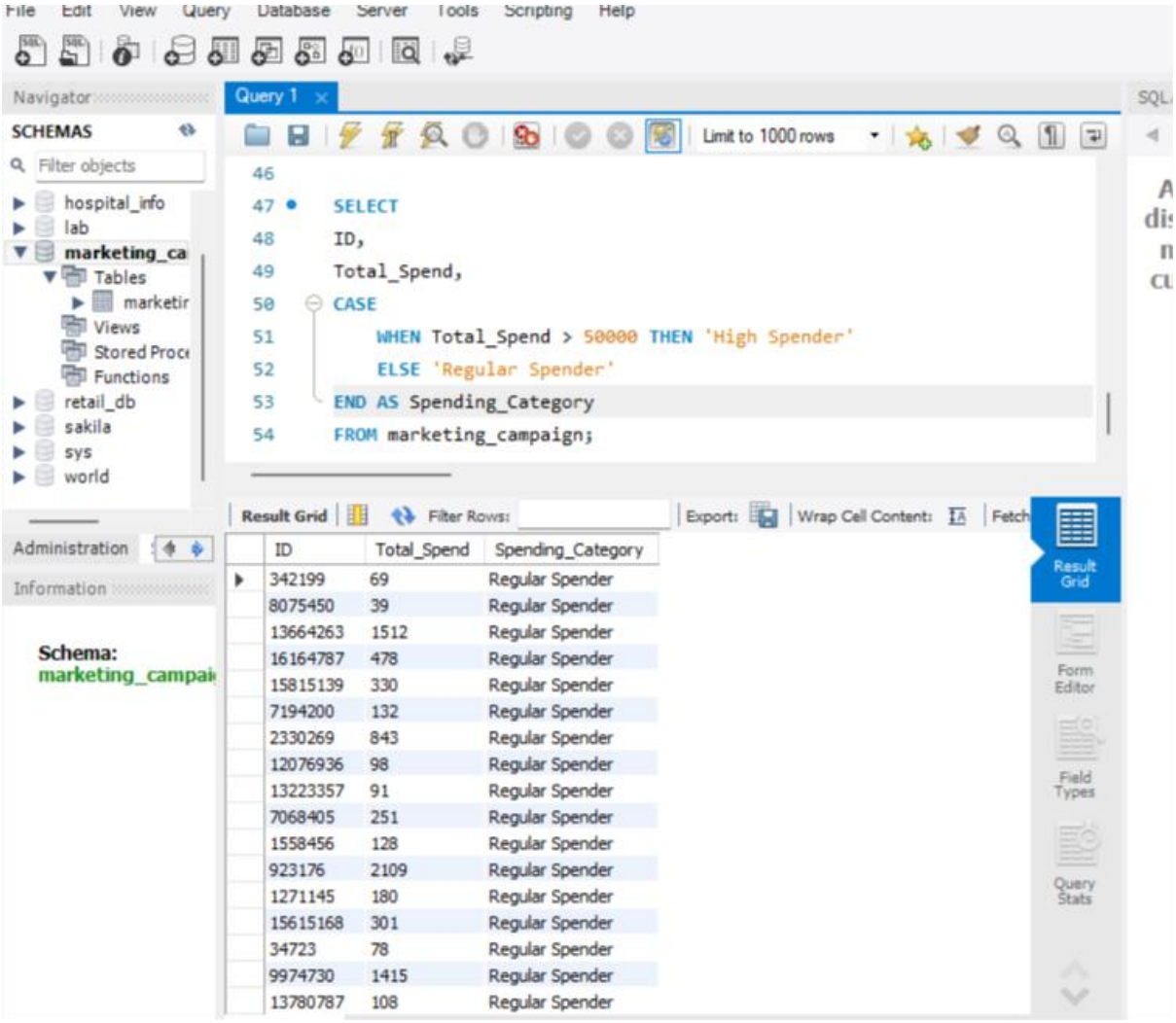
Query 4: High Web Engagement



Explanation

This query classifies customers based on their monthly website visits.

Query 5: Family Customers

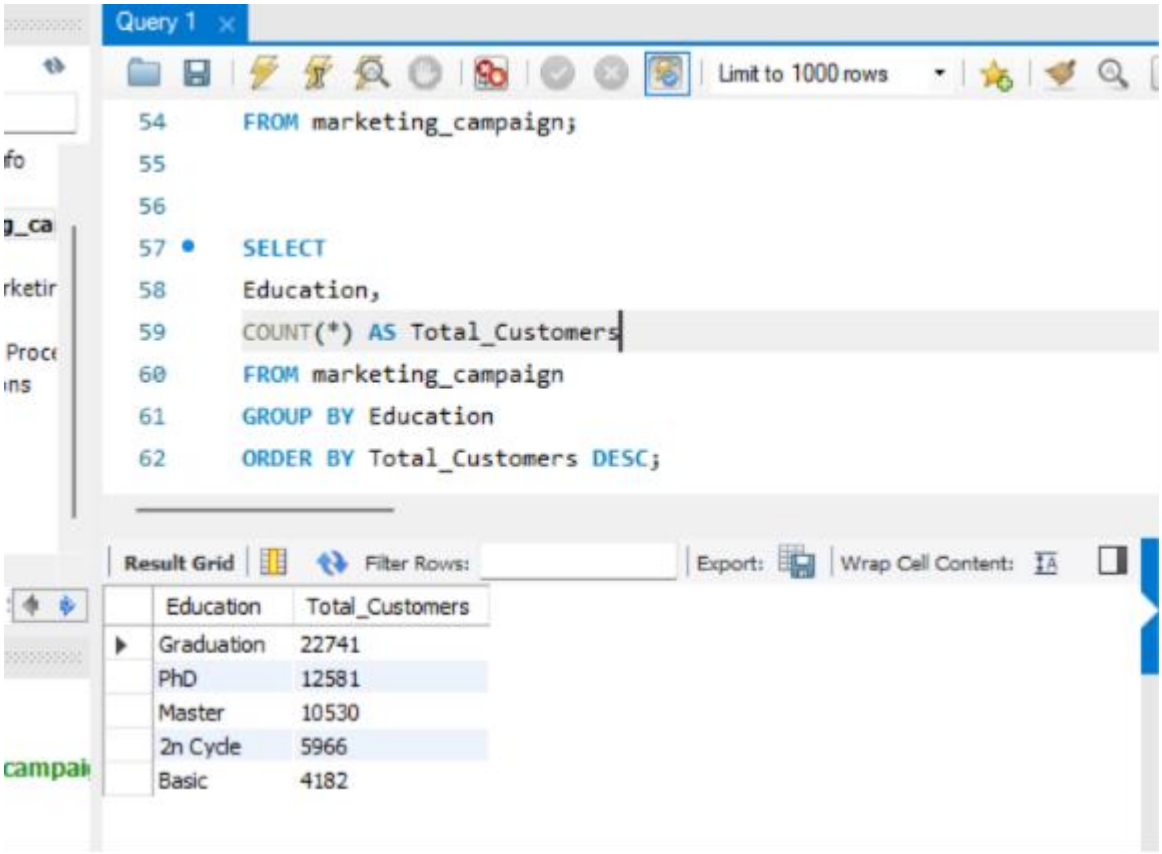


Explanation

This query identifies customers with high spending behavior based on a predefined spending threshold.

Aggregation Queries

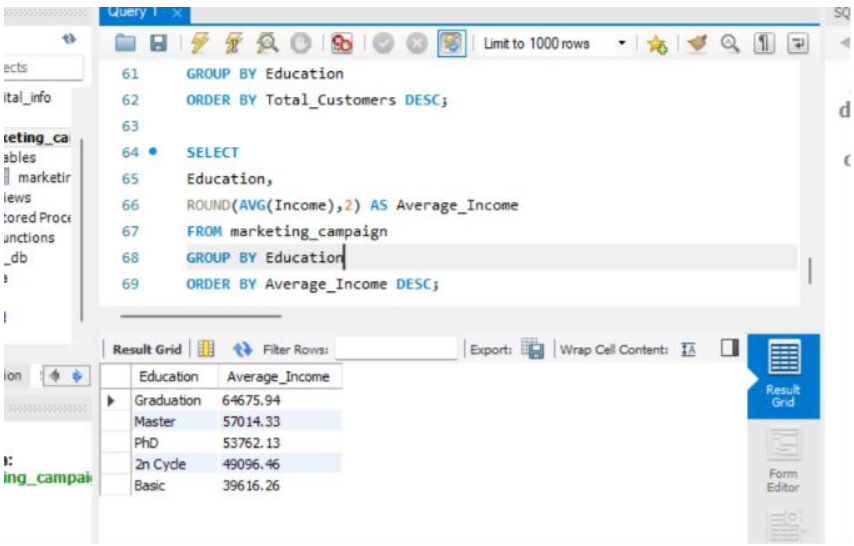
Query 1: Education-wise Customer count



Explanation

This query counts the number of customers in each education category.

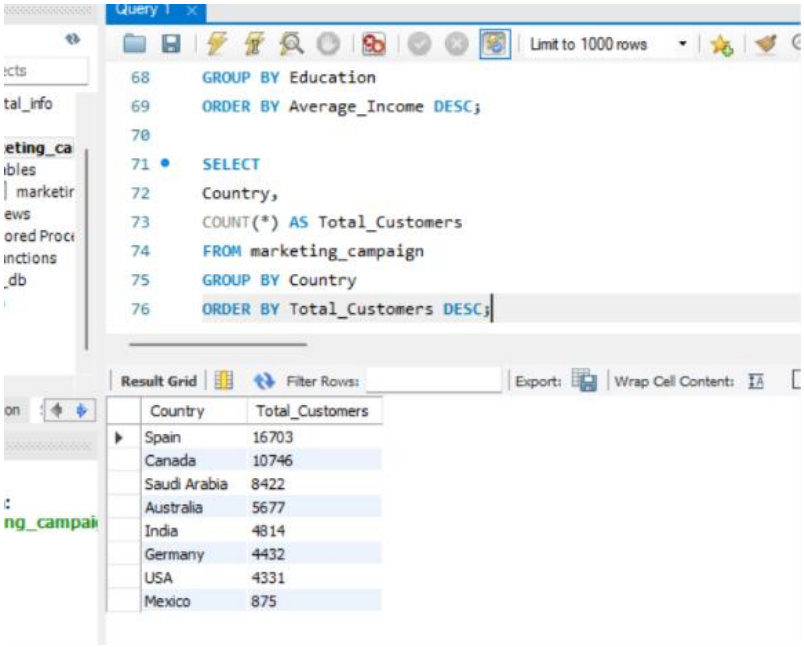
Query 2: Average Income by Education



Explanation

This query calculates the average income for each education level.

Query 3: Country-wise Customer Count



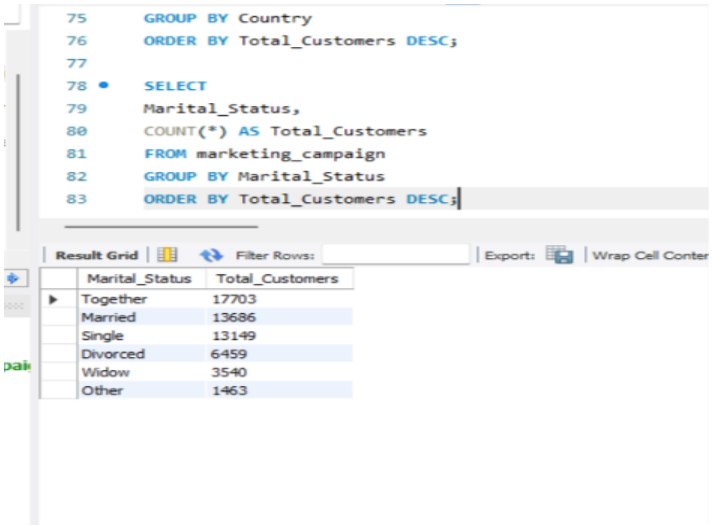
```
68  GROUP BY Education
69  ORDER BY Average_Income DESC;
70
71  SELECT
72  Country,
73  COUNT(*) AS Total_Customers
74  FROM marketing_campaign
75  GROUP BY Country
76  ORDER BY Total_Customers DESC;
```

Country	Total_Customers
Spain	16703
Canada	10746
Saudi Arabia	8422
Australia	5677
India	4814
Germany	4432
USA	4331
Mexico	875

Explanation

This query shows the number of customers from each country.

Query 4: Marital Status Analysis



```
75  GROUP BY Country
76  ORDER BY Total_Customers DESC;
77
78  SELECT
79  Marital_Status,
80  COUNT(*) AS Total_Customers
81  FROM marketing_campaign
82  GROUP BY Marital_Status
83  ORDER BY Total_Customers DESC;
```

Marital_Status	Total_Customers
Together	17703
Married	13686
Single	13149
Divorced	6459
Widow	3540
Other	1463

Explanation

This query shows the distribution of customers based on marital status.

Query 5: Average Spending by Country

```
83    ORDER BY Total_Customers DESC;
84
85    SELECT
86    Country,
87    ROUND(AVG(Total_Spend),2) AS Average_Spending
88    FROM marketing_campaign
89    GROUP BY Country
90    ORDER BY Average_Spending DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	Average_Spending
Australia	827.38
India	723.85
Germany	707.24
USA	663.88
Saudi Arabia	654.57
Mexico	634.14
Canada	585.32
Spain	557.37

Explanation

This query calculates the average customer spending for each country.

Query 6: Campaign Response by Education

```
91
92    SELECT
93    Education,
94    COUNT(*) AS Total_Customers,
95    SUM(Response) AS Responders,
96    ROUND(AVG(Response)*100,2) AS Response_Rate
97    FROM marketing_campaign
98    GROUP BY Education
99    ORDER BY Response_Rate DESC;
```

Limit to 1000 rows

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

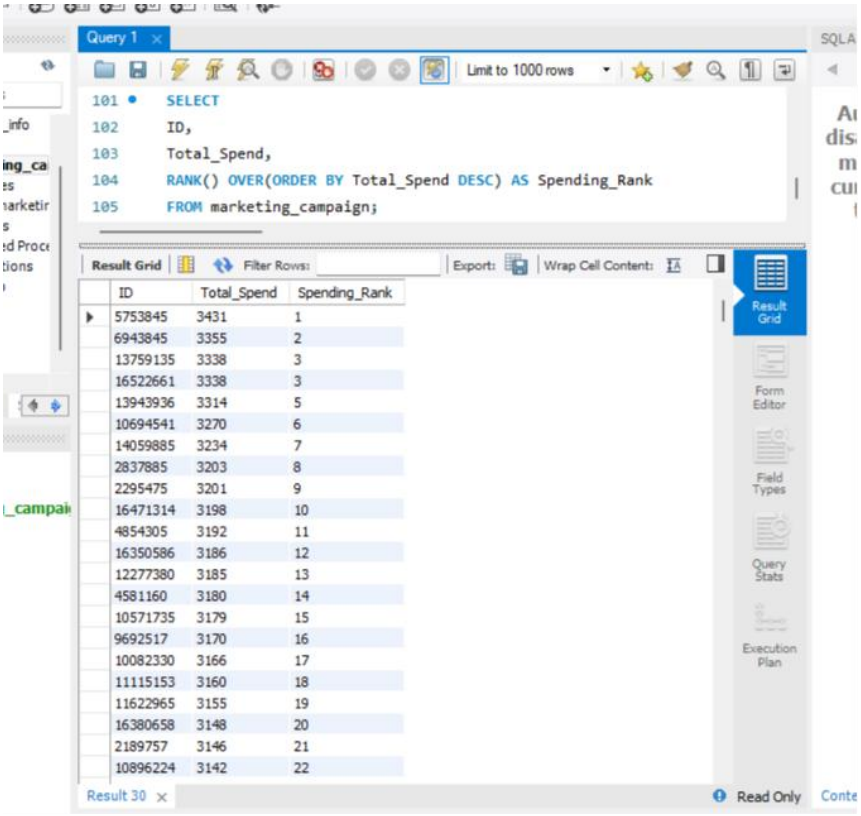
Education	Total_Customers	Responders	Response_Rate
Graduation	22741	3934	17.30
Master	10530	1511	14.35
PhD	12581	1743	13.85
2n Cycle	5966	689	11.55
Basic	4182	388	9.28

Explanation

This query calculates the campaign response rate across different education levels.

Window Functions

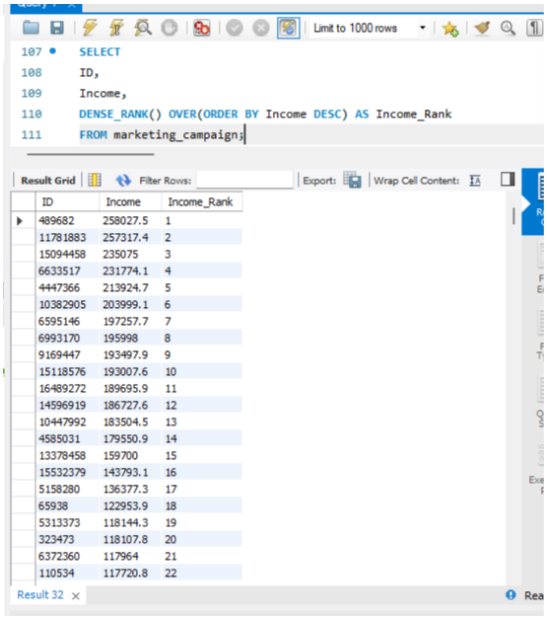
Query 1: Rank Customers by Total Spending



Explanation

This query ranks customers based on their total spending. Customers with the highest spending receive the highest rank. If two customers have the same spending, they receive the same rank.

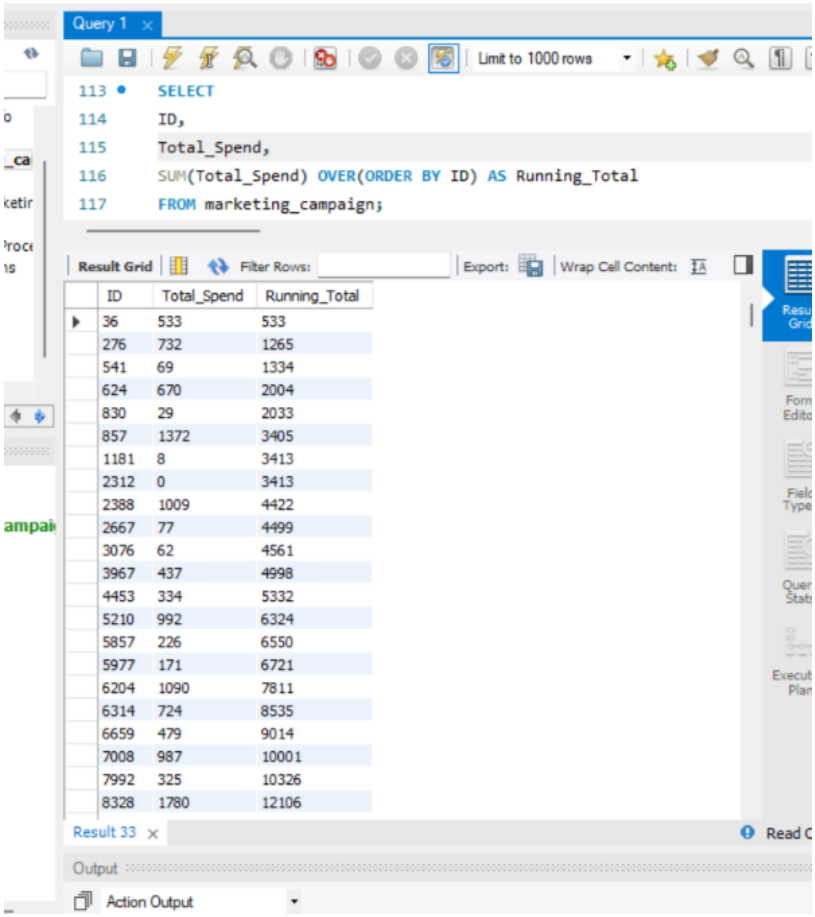
Query 2: Dense Rank by Income



Explanation

This query assigns a dense rank to customers based on their income. Unlike RANK(), DENSE_RANK() does not skip rank numbers when there are ties.

Query 3: Running Total of Spending



Explanation

This query calculates the cumulative spending of customers in ID order using a running total.

Business Insights

- Customers with higher income generally show higher purchasing power.
- High spenders contribute significantly to total revenue.
- Campaign responders represent potential loyal customers for future marketing campaigns.

- Customers with high website engagement are more likely to interact with digital campaigns.
- Family customers can be targeted with bundled or family-oriented offers.
- Education and country-wise analysis helps identify customer segments with better campaign performance.

  This project analyzed customer demographics, purchasing behavior, and campaign responses using Python, SQL, and Streamlit. Data was cleaned and transformed using Pandas, imported into MySQL for SQL-based analysis, and visualized through an interactive dashboard. Business insights generated from KPI analysis, rule-based segmentation, aggregation queries, and window functions can help organizations improve customer targeting and marketing strategies.